

Team Control Number
510105201001220151

Problem Chosen
Wildlife Protection
2026
IM²C
Summary Sheet

Planning Wildlife Protection under Geographic and Resource Constraints

Large reserves need inspections that lead to timely action. We connect conservation priorities, ground response and limited personnel and drones in Etosha, then examine seasonal work and transfer to other parks.

Priorities. We address illegal entry and poaching, harmful fire and habitat damage through a multi-species framework. Six animal groups with usable 2015 survey records and burnable-habitat areas define relative responsibility. The Effective Inspection Completion Rate measures prescribed checks supported by timely ground response.

Allocation. A two-stage linear program first maximizes completion, then minimizes labor across all equally optimal service patterns. With **21240** personnel-hours and **1200** flight-hours available monthly, its **57.26%** optimum uses **5286.11** personnel-hours and **144.34** flight-hours. Response access limits service. The best of three additional-post candidates raises completion to **64.27%**.

Seasonal staffing. Separate fire checks and maintenance of 17 historical boreholes require **57** personnel equivalents in the normal peak month and **62** under the specified drought-maintenance rule, at 120 effective hours per person-month.

Sensitivity. All three allocation rules tie under ample resources. At a 3600-hour labor limit, uniform, demand-based and optimized completion are **16.47%**, **19.87%** and **29.88%**. Reduced drone efficiency restores ground screening; grid size changes the estimated response ceiling.

Evaluation. Independent checks confirm model feasibility and agreement between allocation and inverse-workforce calculations. Historical records and assumed protocols limit operational interpretation; service completion does not establish wildlife survival or sufficient ecological protection.

Transfer. Chitwan and Yellowstone trials rebuild local geography. A historical Yellowstone wolf-use scenario changes priorities and deployment. Chitwan's published geometry and official area remain inconsistent, so its results are conditional. Validate access and task effectiveness before adopting staffing recommendations.

Keywords: wildlife protection; resource allocation; ground response; seasonal workforce; Etosha National Park.

Contents

1	Introduction	3
2	Basic Assumptions and Justifications	3
3	Data Description and Symbol Definitions	4
4	Protection Priorities and a Measurable Standard	4
4.1	Priorities and evidence	4
4.2	What effective inspection completion measures	5
5	Resource Allocation with Ground Teams and Drones	5
5.1	Planning units and resource assumptions	5
5.2	Animal responsibility, habitat and access	6
5.3	Weights, routes and response eligibility	7
5.4	Service capacity and the allocation model	8
5.5	Solution checks and regional allocation	9
5.6	Strategic comparisons and response posts	10
6	Seasonal Service and Workforce Requirements	11
6.1	Target, months and additional tasks	11
6.2	Borehole maintenance as ground-only work	12
6.3	Shared monthly constraints	13
6.4	Forward service and inverse workforce models	13
6.5	Normal-season results and feasibility	14
6.6	Drought work and service-standard sensitivity	15
7	Sensitivity and Scenario Analysis	15
7.1	Screening protocol and grid sensitivity	16
7.2	Weight sensitivity	17
8	Strengths, Limitations and Practical Use	18
8.1	Strengths	18
8.2	Weaknesses and Limitations	18
9	Adaptation to Parks on Other Continents	18
9.1	Shared structure and local data	18
9.2	Conditional trials and expected deployment changes	19
9.3	A local historical-priority trial	20
	Appendix: supplementary results	21
	Letter to IMMC	22
	References	23
	Report on Use of AI	24

Reading guide. Sections 4–9 answer Questions 1–6. The letter uses nontechnical language. References and the AI-use report are included within this document’s 25-page ceiling.

1. Introduction

Large wildlife reserves face a practical protection problem: important locations are spread across a wide area, while personnel, equipment and travel time are limited. Observations are useful only when qualified teams can act on them. Etosha National Park is the starting case, with a reported area of about 22,935 km² and 295 reference employees [1]. We ask how to prioritize inspections, allocate ground and drone work, and maintain a chosen service as seasonal tasks increase.

Our planning units connect historical animal responsibility, burnable habitat and mapped access to explicit checklist demand. A task receives service credit when its prescribed screening is completed and ground staff can arrive within two hours. The resulting Effective Inspection Completion Rate summarizes weighted tasks, rather than wildlife survival or continuous area coverage. Six surveyed animal groups support the quantitative component; missing species and current distributions remain separate evidence needs.

A two-stage linear program maximizes this rate and then minimizes personnel work among equally optimal allocations. A monthly extension maintains the same target while adding fire screening and maintenance of 17 historical boreholes. Inverse workload divided by 120 effective hours gives personnel equivalents. Joint resource, protocol, grid and weight tests examine the assumptions. Trials in Chitwan and Yellowstone rebuild local geography and priorities instead of transferring Etosha's staffing totals.

The baseline rate is 57.26%, using 5286.11 personnel-hours and 144.34 flight-hours. Existing response access limits this result despite spare resources. Normal seasonal work requires 57 personnel equivalents at its peak; the specified drought-maintenance rule requires 62. These are conditional task estimates. Figure 1 organizes the study's data, models, checks and recommendations, showing where numerical verification ends and field calibration remains necessary.

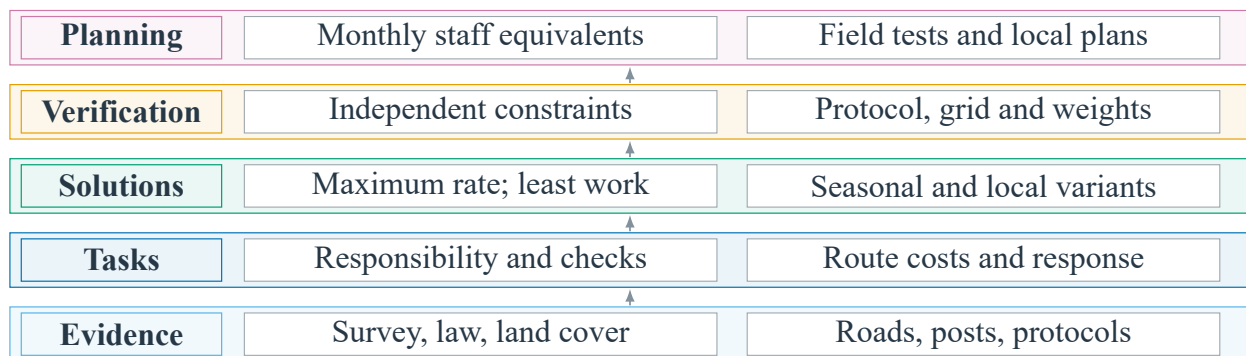


Figure 1: Overall organization of the protection-planning study. The lower layer distinguishes supplied reference values, historical surveys and mapped geography from operating assumptions. The task layer converts animal and habitat responsibility into checklist demand and response eligibility. The solution layer links the maximum-rate allocation to inverse seasonal workforce calculations and local park adaptations. Independent checks then evaluate capacity, units, budgets, service floors and sensitivity, before recommendations are interpreted as conditional workload estimates. For Etosha, the linked calculations give a 57.26% baseline rate and normal and drought-maintenance peaks of 57 and 62 personnel equivalents. Upward arrows represent dependencies, not causal ecological effects or a claim that every input has been field calibrated.

2. Basic Assumptions and Justifications

The following assumptions define a planning scenario rather than verified park operating rules.

Assumption 1: Work is pooled monthly, using 120 effective hours per person, 60% personnel availability and 30 drones.

Justification: Declared limits permit comparable deployment tests; monthly feasibility does not prove a daily roster.

Assumption 2: Inspection demand and within-region responsibility are allocated by area, with equal fire exposure on burnable habitat.

Justification: Fine animal locations and multiyear harmful-fire exposure are unavailable; explicit proxies avoid inventing observations.

Assumption 3: Ground and drone teams complete the same checklist only where suitable; credited service needs ground arrival within two hours and shared teams handle infrequent alerts.

Justification: Observations require follow-up. Task equivalence, service standards and concurrency still need field calibration.

3. Data Description and Symbol Definitions

The problem gives an approximate area of 22,935 km²; our projected boundary covers 22897.10 km². We use 2015 animal surveys, 2021 land cover and a fixed 2026 road snapshot [1, 2, 4, 5], organized into 17 reporting regions and 599 inspection units. A representative location supports route calculations, not continuous area coverage. Question 2 excludes water and rainfall; Question 3 separately adds fire screening and maintenance of 17 historical boreholes without forecasting populations or water-delivery volumes.

Table 1 distinguishes spatial units, completion and the two resource budgets. Service measures checklist work supported by response, independently of ecological outcomes.

Table 1: Symbols used throughout the planning model. Reporting regions aggregate results, inspection units carry local tasks, and months index seasonal work. Areas generate check demand, while normalized weights express responsibility rather than incident probabilities. Completion is credited only where a ground team meets the response deadline. Personnel-hours and flight-hours are separate resources: operating a drone also consumes personnel work. The baseline provides 21240 personnel-hours and 1200 flight-hours monthly, with 2880 personnel-hours reserved once for response readiness. These definitions prevent adding incompatible units or counting response crews separately for every inspection. The reported rate is weighted checklist completion with response support, rather than the fraction of animals surviving or land continuously observed.

Symbol	Definition
i, j, t	Reporting region, inspection unit and month.
A_j, C_j	Unit area and prescribed monthly checks.
w_j, s_j	Relative inspection importance and completion fraction.
P, η	Effective inspection completion rate and target, expressed in percent.
r_j	1 when ground staff can arrive within the time limit; 0 otherwise.
x_j, h_j	Ground screening personnel-hours and drone flight-hours.
H, U, H_0	Personnel limit, flight limit and reserved response work.

4. Protection Priorities and a Measurable Standard

We identify shared threats before assigning regional responsibility. The management goal covers wildlife across habitats; the animal component quantifies the six surveyed groups rather than a complete species inventory.

4.1 Priorities and evidence

Table 2 ranks concerns using threat evidence, potential loss, affected extent and the possibility of intervention. They are management judgments rather than a measured ranking of incident probabilities.

Table 2: Provisional conservation priorities for the Etosha planning case. The ranking uses the supplied threat context, possible losses and opportunities for intervention, rather than measured incident probabilities. Illegal entry and poaching create animal-monitoring and response tasks; harmful fire and habitat disturbance create a separate habitat-monitoring component. Six animal groups with usable 2015 survey records support regional responsibility estimates, while rhinoceros poaching remains a qualitative concern because its count table is unresolved. Grassland, shrubland and forest identify burnable habitat. Disease, human–wildlife conflict and missing priority species require dedicated evidence and safeguards. The two ranked concerns organize the quantified task list without claiming an exhaustive risk inventory or equal vulnerability across species.

Priority	Challenge and evidence
1	Poaching and illegal entry: unauthorized access creates enforcement needs across animal habitats; the supplied rhinoceros-poaching context is one documented example.
2	Harmful wildfire and habitat damage: potentially widespread effects on animals and habitats; distinguish prescribed fire.

The animal component uses blue wildebeest, plains zebra, giraffe, African savanna elephant, gemsbok and ostrich. We retain source quality flags and exclude unresolved count tables. Survey blanks contribute no recorded abundance, rather than confirmed zero populations. These historical records support a deployment demonstration, not current abundance for all wildlife. Rhinoceros poaching remains a qualitative concern; its conflicting counts are not required by this multi-species model. Human–wildlife conflict, disease and species-specific interventions remain outside the quantified task list.

Historical abundance is converted to within-species regional shares, then category preferences determine responsibility. Burnable-habitat area forms a separate component. Equal baseline fire exposure avoids extrapolating a single sample into multiyear harmful-fire risk.

4.2 What effective inspection completion measures

A target receives effective service when prescribed screening is completed and a qualified ground team can arrive within the selected response limit. Let $d_j \in [0, 1]$ describe screening completion and $r_j \in \{0, 1\}$ describe response eligibility. For normalized demand weights w_j , define

$$\begin{aligned}
 g_j^{\text{eff}} &= d_j r_j, \\
 P &= 100 \sum_j w_j g_j^{\text{eff}}, \\
 L &= \sum_j w_j (1 - g_j^{\text{eff}}).
 \end{aligned} \tag{4.1}$$

The allocation uses $s_j \leq r_j$ directly. With fixed weights, larger P means less unmet weighted demand; $P = 57.26$ denotes 57.26%, independently of area coverage or animals protected.

Actual protection needs independent evidence: illegal-loss records with comparable monitoring effort, population surveys, habitat condition and the impact of harmful fires. More detected incidents may reflect better detection. Prescribed burning is not automatically protection failure. A park average cannot certify missing species or replace their dedicated safeguards.

5. Resource Allocation with Ground Teams and Drones

The priorities above determine which inspections receive weight. This section converts their completion requirements into travel, personnel and flight work, then chooses a feasible allocation.

5.1 Planning units and resource assumptions

The 15 survey regions, main salt pan and other survey-external areas form 17 reporting regions $\mathcal{I}$. A 10 km grid intersected with each region produces 599 connected inspection units $\mathcal{J}$. Each has an interior

representative location and support area A_j . The location is a route-planning proxy, not proof of continuous coverage of that area.

Table 3 states the settings. Six sourced locations are assumed response posts; monthly availability does not require a fixed fraction of employees on duty simultaneously.

Table 3: Declared operating assumptions used to construct the Etosha allocation problem. Travel speeds convert mapped routes into task costs and response eligibility; checklist, endurance and openness rules determine feasible monitoring technologies. Each region receives a service floor proportional to its reachable area. A reference workforce of 295 with 60% availability and 120 effective hours supplies 21240 personnel-hours monthly. Thirty proposed drones supply 1200 flight-hours. Six candidate response posts reserve 2880 personnel-hours in total. These settings describe a planning protocol, not verified park staffing, equipment or operating permissions. Sensitivity cases change individual assumptions while retaining the others, so budget shortages can be distinguished from geographic or technological restrictions.

Item	Setting
Travel speed	Road 30, walking 4, drone flight 40 km/h.
Task time / team	Observation 10, preparation 5, dispatch 10 minutes; two people per team.
Service / response	8 checks per 100 km ² /month; ground arrival within 2 hours.
Technology eligibility	At most 0.6 flight-hours/check; regional openness at least 60%.
Access / service floor	Entry-index decay scale 2 hours; reachable-area service floor 15%.
Personnel	295 reference employees; 60% availability; 120 hours/person-month.
Drones	30 aircraft; 2 flight-hours/day for 20 flyable days/month.
Readiness	Six candidate posts, two people each, 8 hours/day for 30 days.

The two resource limits and shared response reservation are

$$\begin{aligned}
 H &= 295(0.60)(120) = 21240 \text{ personnel-hours/month,} \\
 U &= 30(2)(20) = 1200 \text{ flight-hours/month,} \\
 H_0 &= 6(2)(8)(30) = 2880 \text{ personnel-hours/month.}
 \end{aligned} \tag{5.1}$$

The available inspection labor before task allocation is $H - H_0 = 18360$ hours. Response readiness is charged once for the park. Staffing fractions, flight allowances and service standards are explicit planning assumptions; they are not verified operating rules at Etosha.

5.2 Animal responsibility, habitat and access

For species m , let N_{im} be its surveyed abundance in region i , $q_{\ell(m)}$ its category weight, and $\mathcal{I}_S$ the 15 surveyed regions. Let A_i^F denote burnable-habitat area. Define

$$\begin{aligned}
 V_i^A &= \frac{\sum_m q_{\ell(m)} (N_{im} / \sum_{v \in \mathcal{I}_S} N_{vm})}{\sum_m q_{\ell(m)}}, \\
 V_i^F &= \frac{A_i^F}{\sum_v A_v^F}.
 \end{aligned} \tag{5.2}$$

Hunttable, protected and specially protected categories use the team comparison matrix

$$\begin{aligned}
 \mathbf{A} &= \begin{pmatrix} 1 & 1/2 & 1/3 \\ 2 & 1 & 1/2 \\ 3 & 2 & 1 \end{pmatrix}, \\
 \mathbf{q} &= (0.163424, 0.296961, 0.539615)^\top.
 \end{aligned} \tag{5.3}$$

The normalized principal eigenvector gives $\mathbf{q}$; $CR = 0.007933 < 0.1$ checks consistency, not ecological truth. The law supplies categories, not these numerical comparisons [3].

For unit j , the shortest entry time from one of 27 road–boundary intersections is $T_j^E = D_j^E/v_g + d_j^\perp/v_w$. Its access index is $E_j = \exp(-T_j^E/2)$ with time in hours. These are potential access sources, not proven poacher entrances. Figure 2 compares demand with area service. Within a region, $\mu_j = A_j/A_i$ apportions responsibility by area: $R_j^A = V_i^A \mu_j E_j$ and $R_j^F = V_i^F \mu_j$.

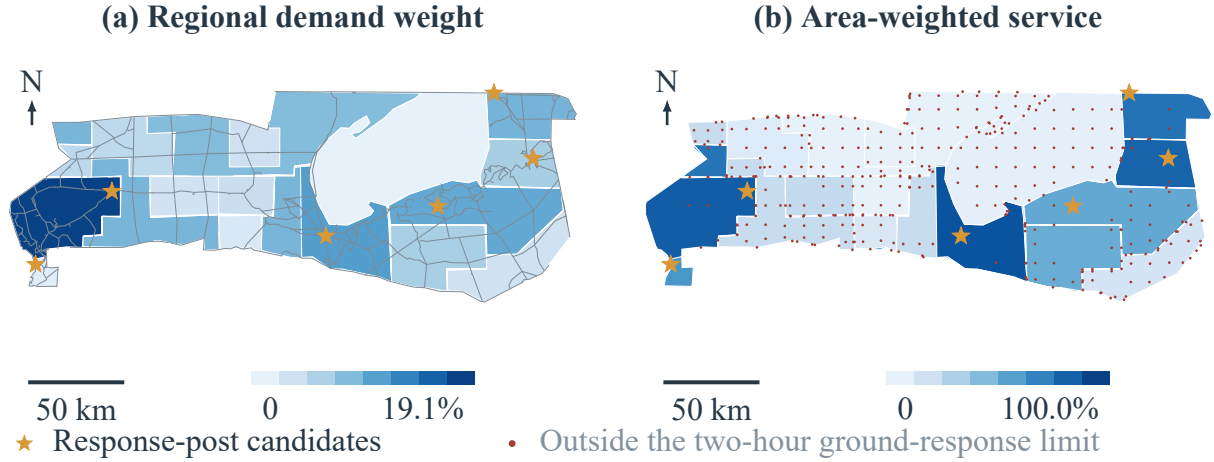


Figure 2: Regional responsibility and base area service in Etosha. The first panel uses six historical animal groups, legal-category preference weights, burnable habitat and an entry-access proxy to construct normalized demand. The second panel aggregates optimized completion by area, using the same 599 inspection units and the two-hour ground-response rule. Stars locate the six assumed response posts; red points identify units that fail that rule. Eligible units cover 34.12% of projected park area, while their priority-weighted completion is 57.26%. The distinction explains why the two panels cannot share a scoring interpretation. Unsurveyed animal values remain unknown, and neither regional shading nor representative-point checks establish continuous observation or measured ecological protection.

Outside-survey animal responsibility remains unknown; zero measured coefficients and zero salt-pan fuel do not imply zero ecological value.

5.3 Weights, routes and response eligibility

We estimate the two component weights over the common 15-region scope. First aggregate $X_{ik} = \sum_{j \in \mathcal{J}_i} R_j^k$, then calculate

$$\begin{aligned} p_{ik} &= \frac{X_{ik}}{\sum_v X_{vk}}, \\ e_k &= -\frac{\sum_i p_{ik} \ln p_{ik}}{\ln 15}, \\ \alpha_k &= \frac{1 - e_k}{(1 - e_A) + (1 - e_F)}. \end{aligned} \quad (5.4)$$

With $0 \ln 0 = 0$, $e_A = 0.881596$ and $e_F = 0.934989$ give $\alpha_A = 0.645551$ and $\alpha_F = 0.354449$. Entropy weights express spatial differences in the chosen data, not observed loss shares. Each component is separately normalized over its observed scope, then $w_j = \alpha_A w_j^A + \alpha_F w_j^F$ and $W_i = \sum_{j \in \mathcal{J}_i} w_j$. Thus $\sum_j w_j = 1$.

The directed road graph retains original connections and one-way restrictions. Outbound and return distances are calculated separately. A post qualifies only if its round trip is finite and

$$\begin{aligned} t_d + d_{bj}^+/v_g + d_j^\perp/v_w &\leq T_{\max}, \\ r_j &= \mathbf{1}\{\text{at least one qualifying post}\}. \end{aligned} \quad (5.5)$$

We choose the qualifying post with the shortest road round trip $D_j = d_{bj}^+ + d_{jb}^-$. Response eligibility concerns the outbound journey; task cost also includes the return. Authorized use of candidate roads and straight-line off-road travel are assumptions requiring field checks.

$$\begin{aligned}
a_j &= n_g(D_j/v_g + 2d_j^\perp/v_w + t_o + t_p), \\
b_j &= 2d_j^\perp/v_u + t_o, \\
o_j &= n_u(D_j/v_g + t_p + b_j), \\
\gamma_j &= o_j/b_j.
\end{aligned} \tag{5.6}$$

Ground cost a_j and drone-support cost o_j are personnel-hours per check; b_j is flight-hours per check. Drone use h_j therefore consumes $\gamma_j h_j$ personnel-hours. Only response-eligible units meeting endurance and openness conditions enter $\mathcal{J}_U$. Figure 3 compares work for one example check.

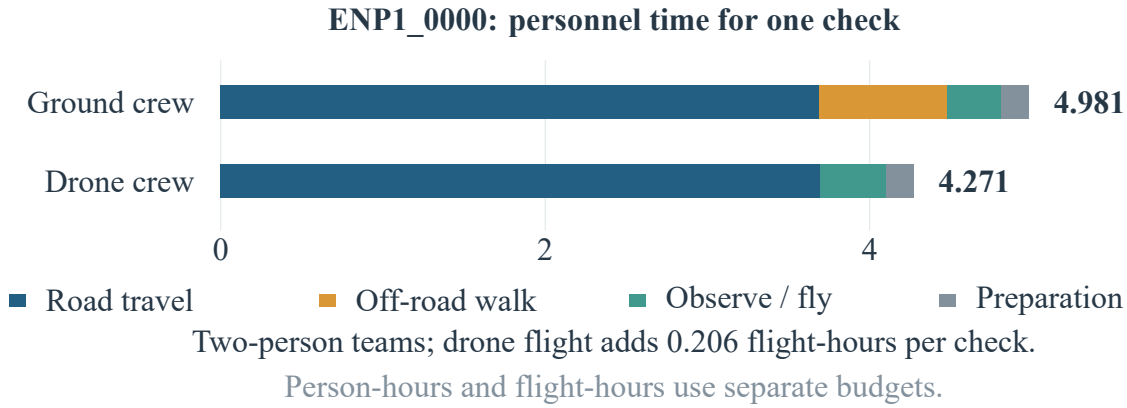


Figure 3: Personnel work required for one checklist inspection at ENP1_0000. Costs are calculated from the saved route, off-road distance, two-person crews, preparation and observation times. A ground inspection requires 4.981 personnel-hours; the drone-assisted alternative requires 4.271 personnel-hours and separately consumes 0.206 flight-hours. Both retain road transport and human involvement. The colored segments distinguish road travel, off-road walking, observation or flight, and preparation, making the assumed substitution explicit. Savings arise from replacing slow off-road walking in a technically suitable location, not from removing operators or responders. These quantities are modeled per-check costs rather than observed patrol durations, and checklist equivalence must be tested before operational deployment.

5.4 Service capacity and the allocation model

Decision variables are ground screening hours x_j , drone flight-hours h_j , and effective completion s_j . The prescribed monthly checks are based on area:

$$\begin{aligned}
C_j &= 8A_j/100, \\
\sum_j C_j &= 8 \sum_j A_j/100.
\end{aligned} \tag{5.7}$$

Areas are in km^2 , giving 1831.77 prescribed checks per month. Refining the grid redistributes area without creating additional total demand. A completed check is a task at a representative location. Fractional checks describe planning intensity and require integer dispatches for implementation.

Define capacity coefficients $g_j = 1/a_j$ for eligible ground tasks, otherwise zero; $u_j = 1/b_j$ for eligible drone tasks, otherwise zero. Then $C_j s_j \leq g_j x_j + u_j h_j$. This avoids dividing by undefined costs and permits different checks to be shared between technologies without double-counting the same inspection.

For each region, $Q_i = \sum_{j \in \mathcal{J}_i} \mu_j r_j$ is reachable area share and $\bar{s}_i = \sum_{j \in \mathcal{J}_i} \mu_j s_j$ is area-weighted completion. A common floor $\bar{s}_i \geq 0.15Q_i$ protects basic service within the reachable part. If only 20% of a region is reachable, the full-region floor is 3%, not 15%.

$$\begin{aligned}
& \max_{\mathbf{x}, \mathbf{h}, \mathbf{s}} \quad 100 \sum_j w_j s_j \\
& \text{s.t.} \quad \left\{ \begin{array}{ll}
C_j s_j \leq g_j x_j + u_j h_j, & j \in \mathcal{J}, \\
\sum_j x_j + \sum_{j \in \mathcal{J}_U} \gamma_j h_j \leq H - H_0, & \\
\sum_j h_j \leq U, & \\
\sum_{j \in \mathcal{J}_i} \mu_j s_j \geq 0.15 Q_i, & i \in \mathcal{I}, \\
0 \leq s_j \leq r_j, & j \in \mathcal{J}, \\
x_j, h_j \geq 0, & j \in \mathcal{J}, \\
x_j = 0, & r_j = 0, \\
h_j = 0, & j \notin \mathcal{J}_U.
\end{array} \right. \tag{5.8}
\end{aligned}$$

The fixed coefficients make this a linear program. Completion is capped at one so excess inspections cannot raise the completion rate above 100%. Human and flight budgets have different units and are linked through operator labor, not added together.

First maximize completion to obtain P^* . Then minimize $H_M = \sum_j x_j + \sum_{j \in \mathcal{J}_U} \gamma_j h_j$ under the same constraints and $100 \sum_j w_j s_j \geq P^*$. All service fractions remain decision variables. This gives the least labor among optimal-rate allocations in the stated model. At the geographic ceiling, positive-weight reachable units must be fully served; zero-weight units still satisfy the regional floor.

5.5 Solution checks and regional allocation

HiGHS solves 1797 variables; independent checks verify capacity, budgets, floors and prohibited allocations. Figure 4 maps all regions, and Table 4 supplies their numerical totals. ENP9 has no qualified response unit, while ENP11 combines substantial demand with almost no reachable area.

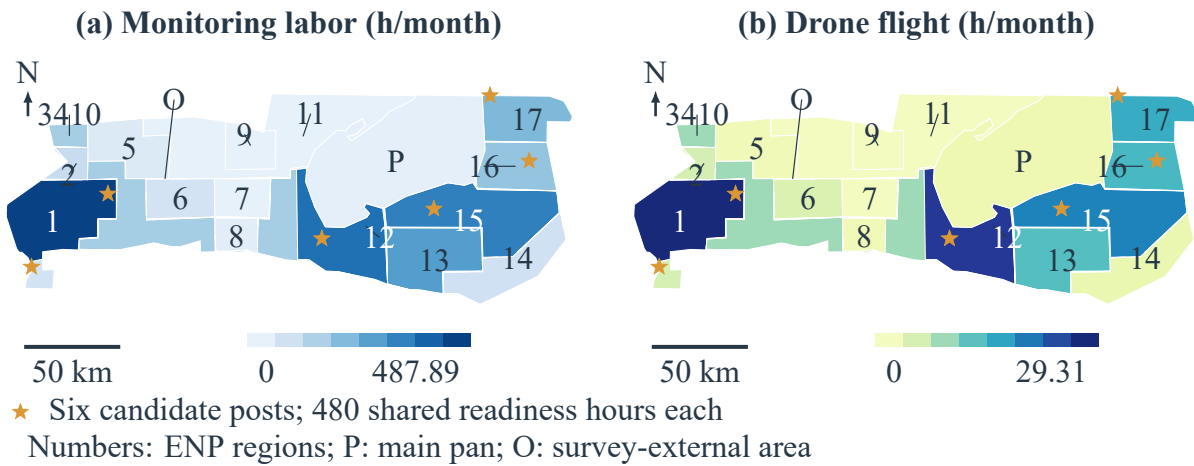


Figure 4: Spatial distribution of the minimum-labor optimal-rate allocation across all 17 reporting regions. Panel (a) maps monitoring and drone-support personnel work; panel (b) maps drone flight-hours. Each scale is independently normalized, so matching colors across panels do not imply equal resource quantities. Numeric labels identify ENP regions, including the combined 3410 unit; P denotes the main pan and O denotes survey-external areas. Leaders identify displaced labels. ENP1 has the largest allocations, 487.89 personnel-hours and 29.31 flight-hours. Six stars mark assumed response posts, each reserving 480 personnel-hours outside regional shading. The maps show pooled monthly workloads under fixed sites, rather than aircraft locations, continuous coverage or measured ecological outcomes.

Table 4: Regional demand and the minimum-labor allocation at the optimal Etosha rate. Demand uses conservation-priority weights, whereas reach and service are area-weighted fractions within each reporting region. Labor includes ground screening and drone support, and excludes the shared 2880 readiness hours. The global second stage preserves the 57.26% optimum while allowing all completion fractions to change. ENP1 receives 487.89 monitoring-support hours and ENP12 receives 385.64; ENP9 receives no credited service because no unit qualifies for timely response. The salt pan retains its regional service floor despite a zero measured-component weight. Unknown animal values are not imputed zeros, and monthly regional totals do not specify integer employees or aircraft stationed locally.

Region	Demand (%)	Reach (%)	Service (%)	Labor (h)	Flight (h)
ENP1	19.06	87.87	87.87	487.89	29.31
ENP12	11.02	92.49	92.49	385.64	26.87
ENP15	9.96	52.33	52.33	352.52	19.90
ENP17	8.84	77.60	77.60	214.11	15.58
ENP3410	8.71	19.21	19.21	155.98	8.38
ENP11	8.05	0.02	0.02	0.26	0.01
ENP16	5.93	84.63	84.63	183.34	14.00
ENP13	5.88	49.87	49.87	282.75	13.08
ENP2	4.50	76.71	76.71	92.86	4.44
ENP5	4.39	5.94	5.94	30.79	0.80
ENP7	3.31	0.19	0.19	0.52	0.02
ENP9	2.90	0.00	0.00	0.00	0.00
ENP14	2.63	11.67	11.67	72.40	2.06
ENP6	2.55	20.57	20.57	60.50	3.62
ENP8	1.54	8.55	8.55	14.93	0.75
Other*	0.72	61.81	61.81	62.52	4.19
Salt pan*	0.00	9.05	1.36	9.11	1.35

* Animal values unknown. Regional labor excludes shared response readiness.

5.6 Strategic comparisons and response posts

Since $w_j \geq 0$ and $s_j \leq r_j$, the geographic upper bound on effective inspection completion is

$$P \leq P_{\text{geo}} = 100 \sum_j w_j r_j. \quad (5.9)$$

The base reaches this bound with 15953.89 personnel-hours and 1055.66 flight-hours unused. Of 599 units, 216 meet the response standard, covering 34.12% of projected area. The 57.26% effective inspection completion rate uses priority weights; the 34.12% figure measures response-eligible area.

We compare the optimized allocation with uniform completion over reachable units and completion proportional to regional demand density W_i/A_i . Costs, resource limits and floors are identical.

Table 5 compares equal-budget rules before considering new response sites.

Table 5 locates the advantage under scarcity. Removing drones still gives 57.26%, at 6084.98 personnel-hours. Equal crews and checklist times make suitable drone use save off-road walking rather than eliminate personnel.

Table 6 compares additional posts, each adding 480 readiness hours within the same total budgets. Routes and response eligibility are recalculated, with demand weights fixed.

Table 5: Completion rates for three allocation rules under identical tasks, weights, response gates and resource constraints. Uniform service equalizes completion among eligible units; demand-based service follows regional demand density; optimization chooses completion and technologies jointly. All three achieve the 57.26% geographic ceiling at the 21240-hour baseline and after a 40% labor reduction. A separate 3600-hour stress budget includes 2880 readiness hours and retains 1200 flight-hours. Its rates are 16.47%, 19.87% and 29.88%, respectively. The contrast identifies an optimization advantage under scarcity without inventing a baseline improvement. Each plan satisfies the same regional floors, and the comparison concerns checklist service rather than observed incident reduction or animal survival.

Budget	Labor limit	Uniform	Demand-based	Optimized
Base	21240	57.26	57.26	57.26
Personnel -40%	12744	57.26	57.26	57.26
Tight budget	3600	16.47	19.87	29.88

Table 6: Additional response-post candidates tested within the original total resource budgets. Each candidate adds 480 readiness personnel-hours, and routes, response eligibility and inspection costs are rebuilt while evaluation weights remain fixed. The original six-post configuration achieves 57.26% using 5286.11 personnel-hours. Adding the ENP5 candidate achieves 64.27% using 6202.08 personnel-hours, a 7.01 percentage-point gain. The other two candidates provide smaller rate gains with different workload requirements. This enumeration isolates the effect of response access rather than increasing the overall labor allowance. ENP5 is best among three tested locations, not a globally optimized station network. Mapped access, permissions, incident concurrency and facilities require verification before a site recommendation becomes operational.

Layout	Readiness (h)	Rate (%)	Total labor (h)	Flight (h)
Original six posts	2880	57.26	5286.11	144.34
Add ENP11	3360	58.86	5888.64	154.05
Add ENP3410	3360	58.95	5825.11	151.03
Add ENP5	3360	64.27	6202.08	181.77

ENP5 is best among three candidates, adding 7.01 percentage points; it is not a global siting optimum. Verify facilities and access before adoption.

6. Seasonal Service and Workforce Requirements

Question 3 asks how service changes through time and how many people are needed to maintain a chosen level. We extend the basic inspections with seasonal work, then convert minimum workload into personnel requirements.

6.1 Target, months and additional tasks

We use the same weights, inspection standard, routes, response locations and technical eligibility as Question 2. The target is its achieved optimum of 57.26%, using the unrounded percentage value $\eta = 57.26016128$; its ecological sufficiency is not established by being mathematically optimal.

Each planning month has 30 days and 120 effective personnel-hours per person. Reference availability is 177 people and 1200 monthly flight-hours. Assignments may change between months. Existing equipment, operators and response staff cannot be charged twice for concurrent tasks.

The dry-season classification is May–October in the cited study; the official fire strategy distinguishes May–July and August–November operating windows [9, 6]. These sources support the timing, not the numerical task frequencies. Table 7 gives the assumed frequencies.

Table 7: Medium seasonal service rules for the monthly workforce model. The cited seasonal classifications establish the operating windows, while numerical visit frequencies and extra checklist requirements are team planning assumptions. Regular maintenance visits occur twice per point in wet months and four times in dry months. Extra fire screening increases from zero to two and then four checks per 100 square kilometers of burnable habitat. November combines wet maintenance with the late-fire rule. Each month represents 30 planning days, and each available worker supplies 120 effective hours. These additional tasks share the baseline personnel and flight resources; they do not redefine the basic inspection score or predict animal abundance, rainfall, fire occurrence or water-delivery volumes.

Months	Water visits/point	Extra fire checks*
Jan–Apr, Dec	2	0
May–Jul	4	2
Aug–Oct	4	4
Nov	2	4

*Checks per 100 km² of burnable habitat per month; on-site maintenance is 0.5 hour/visit in normal months.

Extra fire checks are distinct tasks added to the base checklist. Let A_j^{fuel} be the unit's support area multiplied by its region's observed burnable-habitat fraction. Then

$$D_{jt} = k_t \frac{A_j^{\text{fuel}}}{100} r_j. \quad (6.1)$$

The same inspection protocol and cost coefficients apply. These are screening tasks, not firefighting or prescribed burning. The factor r_j restricts credited tasks to the existing response range; excluded demand is separately recorded.

At the late-fire rate, 275.76 additional checks are supported inside that range, while 428.45 fall outside it. Adding personnel to the same six-post layout cannot guarantee those excluded tasks. We retain this gap rather than claiming park-wide fire coverage.

6.2 Borehole maintenance as ground-only work

A published 2013 field table contains 42 water-point records; 17 are classified as boreholes [7]. We use those 17 mapped points as a historical maintenance sample. It is not a complete current operational inventory. Natural springs are not assigned pumping-equipment work. Ministry records support solar-powered borehole equipment and maintenance as relevant activities [8].

Each visit includes a directed road round trip from a candidate post, any required walking, preparation and on-site inspection. We assume two people travel together. Let T_ℓ^{rt} be elapsed round-trip travel time, f_t visits per point and τ_t on-site time. Monthly personnel work is

$$\begin{aligned} a_{\ell t}^W &= 2(T_\ell^{\text{rt}} + \tau_t + 1/12), \\ W_t &= f_t \sum_{\ell \in \mathcal{L}} a_{\ell t}^W. \end{aligned} \quad (6.2)$$

The 1/12 preparation term is five minutes. Physical maintenance remains ground work. Table 8 compares the three rules.

Summed round-trip travel is 42.87565 elapsed hours. The regular dry-month workload is

$$W_{\text{dry}} = 4(2) [42.87565 + 17(0.5 + 1/12)] = 422.34 \text{ personnel-hours}. \quad (6.3)$$

Major repairs, unlisted points and route-sharing need updated records. The longest visit takes about 6.39 elapsed hours; a single task fitting an eight-hour day does not prove a roster.

Table 8: Maintenance workloads for the 17 boreholes documented in the historical field sample. Every modeled visit uses a two-person crew, a directed round trip, five minutes of preparation and the specified on-site work. Summed elapsed round-trip travel is 42.87565 hours. Two regular visits produce 211.17 personnel-hours monthly; four produce 422.34. The drought-maintenance scenario assumes eight visits and one on-site hour, producing 980.68 personnel-hours. These differences result from service rules rather than measured faults or inferred water shortages. Independent round trips claim no combined-route savings. The sample is not a complete inventory of currently functioning facilities, and drones cannot replace physical equipment maintenance, major repairs or unmeasured emergency work.

Rule	Visits/point	On-site hours	Monthly labor
Wet regular	2	0.5	211.17
Dry regular	4	0.5	422.34
Drought pressure	8	1.0	980.68

6.3 Shared monthly constraints

For each unit and month, (x_{jt}, h_{jt}, s_{jt}) serve the base inspection protocol. Extra fire work uses ground hours y_{jt} and flight-hours v_{jt} . These are separate task quantities with shared personnel and flight budgets. Where a technology is unavailable, its capacity term and investment are zero.

$$\begin{aligned}
 C_j s_{jt} &\leq g_j x_{jt} + u_j h_{jt}, \\
 D_{jt} &\leq g_j y_{jt} + u_j v_{jt}, \\
 \sum_j (h_{jt} + v_{jt}) &\leq U,
 \end{aligned} \tag{6.4}$$

$$H_t = H_0 + W_t + \sum_j (x_{jt} + y_{jt}) + \sum_{j \in \mathcal{J}_U} \gamma_j (h_{jt} + v_{jt}).$$

The remaining conditions retain $0 \leq s_{jt} \leq r_j$, nonnegative investments, technology eligibility and the regional floor

$$\sum_{j \in \mathcal{J}_i} \mu_j s_{jt} \geq 0.15 \sum_{j \in \mathcal{J}_i} \mu_j r_j. \tag{6.5}$$

Only basic checks enter $P_t = 100 \sum_j w_j s_{jt}$. Required seasonal tasks use the same budgets without changing that score. Readiness is charged once; monthly feasibility still requires date, qualification and concurrent-alert checks.

6.4 Forward service and inverse workforce models

Two uses of the same feasible task set answer different questions. With N available people, the forward model finds the best monthly service. With a fixed service target, the inverse model finds the least total personnel work:

$$\begin{aligned}
 P_t^*(N, U) &= \max_{\mathbf{z}_t \in \mathcal{F}_t(U), H_t \leq 120N} 100 \sum_j w_j s_{jt}, \\
 H_t^{\min}(\eta, U) &= \min_{\mathbf{z}_t \in \mathcal{F}_t(U), P_t \geq \eta} H_t.
 \end{aligned} \tag{6.6}$$

Here $\mathbf{z}_t$ collects all five variable blocks and $\mathcal{F}_t(U)$ contains the shared constraints. The inverse model removes the reference labor ceiling so that a shortage can be expressed as a required workforce instead of only returning infeasibility. As in revised Question 2, it allows service fractions to vary among equal-rate allocations.

$$N_t = \left\lceil \frac{H_t^{\min}}{120} \right\rceil, \quad (6.7)$$

$$N_{\text{year}} = \max_{t=1, \dots, 12} N_t.$$

Figure 5 separates monthly work. The maximum, rather than the sum, estimates a constant workforce meeting the busiest month's pooled workload. It does not certify qualifications, exact simultaneous coverage or the completeness of the task inventory.

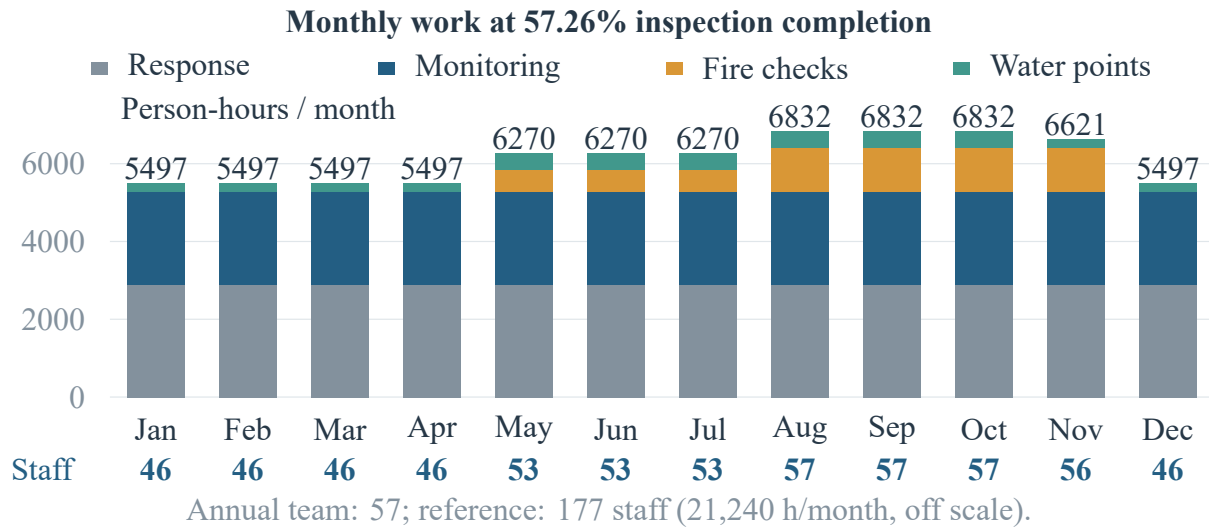


Figure 5: Minimum monthly work required to preserve the 57.26% baseline inspection target under the medium seasonal rules. The stacked bars separate shared response readiness, basic inspection support, extra fire checks and maintenance of 17 historical boreholes. August through October are the busiest months at 6831.83 personnel-hours, requiring 57 available personnel equivalents after division by 120 and upward rounding. The reference capacity of 177 personnel is outside the plotted workload scale. Seasonal increases use spare modeled resources while the response-limited baseline score remains unchanged. These pooled monthly quantities explain strategic workforce needs; they do not certify an executable daily roster, personnel qualifications, simultaneous alert coverage or the completeness of the park's real task inventory.

With seasonal tasks removed, the independently constructed inverse model returns 5286.11 personnel-hours, matching revised Question 2. This cross-check confirms that both chapters use the same optimal-rate, minimum-labor objective.

6.5 Normal-season results and feasibility

Table 9 reports that all months maintain the target with the 177-person reference and 1200 flight-hours. The least-work solutions need fewer personnel equivalents because only specified tasks and the existing response range are modeled.

The peak occurs in August–October. Its work separates into four parts:

$$H_{\text{peak}}^{\min} = 2880 + 2406.11 + 1123.38 + 422.34 = 6831.83 \text{ personnel-hours.} \quad (6.8)$$

These are readiness, base inspection labor, extra fire inspection labor and borehole maintenance. The flight requirement is 210.61 hours, well below 1200. Rounding $H_{\text{peak}}^{\min}/120$ gives 57 people. A check with 56 people returns 56.73, below the target, under the same monthly constraints.

Table 9: Minimum work and personnel equivalents for the four medium-rule seasonal periods. Each row preserves the same unrounded 57.26016128 baseline target and uses the shared 1200 flight-hour allowance. January through April and December require 5497.28 personnel-hours and 46 equivalents. May through July require 53; August through October require 57; November requires 56. Flight work peaks at 210.61 hours. Fixed reference availability of 177 personnel maintains the target in all periods because its capacity exceeds the included workload. The annual fixed team is the maximum monthly requirement rather than the sum of monthly headcounts. These estimates retain the existing response range and assumed checklist, so omitted specialist, night or concurrent incident work may change actual staffing needs.

Months	Personnel-hours	People	Flight-hours	Rate at 177 staff (%)
Jan–Apr, Dec	5497.28	46	144.34	57.26
May–Jul	6270.14	53	177.48	57.26
Aug–Oct	6831.83	57	210.61	57.26
Nov	6620.66	56	210.61	57.26

This is a task-specific monthly minimum, not an operational staffing recommendation. Specialist, night and simultaneous incident work are omitted.

Independent task-capacity, qualification, budget and floor checks verify the saved plans. Field costs, permissions and checklist effectiveness require separate calibration.

Extra seasonal work uses spare capacity without expanding the fixed response range. Raising the basic target requires improved access or deployment.

6.6 Drought work and service-standard sensitivity

The drought-maintenance case doubles dry-month visits and on-site time. Its peak is 7390.17 personnel-hours, or 62 equivalents, with 210.61 flight-hours. Table 10 compares alternative rules.

Table 10: Workforce requirements under low, medium and high assumed seasonal service rules. The alternatives change maintenance visits and extra fire screening together, and each is solved for the same baseline inspection target. Normal peak personnel equivalents are 51, 57 and 71; the specified drought-maintenance rules raise them to 53, 62 and 77. Corresponding personnel-hours include readiness, screening support and physical maintenance rather than flight time alone. The table therefore measures the consequence of prescribed work, not a statistical confidence interval or a forecast of future drought. A lower required headcount under a relaxed checklist cannot be called unchanged ecological protection at lower cost. Selecting an operational rule requires task-duration, maintenance-status and outcome evidence.

Rule	Normal people	Normal hours	Drought people	Drought hours
Low	51	6041.97	53	6270.14
Medium	57	6831.83	62	7390.17
High	71	8513.55	77	9207.89

Different seasonal rules alter the prescribed service. Fewer personnel under a relaxed rule do not establish equal ecological protection.

7. Sensitivity and Scenario Analysis

Figure 6 tests joint budgets. Reducing baseline personnel-hours by 40% still gives 57.26% effective inspection completion. Lower resource budgets are tested with weights and tasks fixed. All 127 feasible saved solutions pass independent checks; four infeasible cases are excluded rather than plotted as zero completion.

Mid-level seasonal peak: joint personnel and flight capacity						
Staff	0	80	160	200	240	1200
45	20.3%	33.8%	37.2%	37.2%	37.2%	37.2%
55	44.7%	52.0%	55.2%	55.7%	55.7%	55.7%
57	47.7%	54.1%	57.0%	57.3%	57.3%	57.3%
62	53.6%	57.3%	57.3%	57.3%	57.3%	57.3%
67	57.3%	57.3%	57.3%	57.3%	57.3%	57.3%
177	57.3%	57.3%	57.3%	57.3%	57.3%	57.3%

Columns: flight-hours/month; cells: effective inspection completion (%).

Target 57.26%; 177 staff and 1200 flight-hours are the reference.

Figure 6: Peak-month effective inspection completion under joint personnel and drone-flight limits. Each cell is independently reoptimized for the medium seasonal workload, including mandatory maintenance and extra fire screening. Rows specify available personnel at 120 effective hours each, and columns specify monthly flight-hours. A 57-person team with 200 flight-hours maintains the 57.26% target, whereas the same team with 160 hours reaches 56.97%. With no drone capacity, maintaining the target requires 67 people. This interaction prevents extrapolating one-resource tests into a joint shortage. Cell values describe the stated model and response layout; they are not detection probabilities, confidence limits, a real roster or a claim that all proposed aircraft can fly simultaneously.

At the medium peak, 57 people with 200 flight-hours maintain the target; 57 with 160 hours reach 56.97%. With no drones, the target needs 67 people. With 59 people and 120 flight-hours, reallocation reaches 57.24%, compared with 57.14% under the retained spatial service rule. Under the specified drought-maintenance pressure at the same low budgets, the gain from reallocation is 6.83 percentage points, but the target is still missed.

For the same 68 borehole visits and 422.34 monthly personnel-hours, staggered dates reduce the maximum gap from 27 to 8 days and the daily water-work peak from 105.58 to 15.48 hours. This water-only schedule is not a full roster or proof of a strict 7.5-day rule.

At road speeds of 20 and 40 km/h, the completion-rate ceilings are 35.76% and 64.69%. Extending response to three hours gives 80.64% under a changed service standard. These are conditional scenarios, not measured speed ranges or event probabilities.

7.1 Screening protocol and grid sensitivity

Let e be drone screening capacity relative to the baseline checklist and p_g the required ground-check share. Replace task capacity by $C_{js_j} \leq x_j/a_j + eh_j/b_j$ and, where required, impose $x_j/a_j \geq p_g C_{js_j}$. The baseline has $e = 1$ and $p_g = 0$. Table 11 tests these assumed protocols, rather than measured detection probabilities.

Table 11: Sensitivity of technology choice to assumed inspection protocols under baseline budgets. Drone efficiency scales completed checklist capacity relative to baseline, and a ground minimum requires an explicit share of completed checks to use ground screening. All tested protocols retain the 57.26% geographic ceiling, but workloads change. At 75% drone efficiency, ground screening returns at 1731.55 personnel-hours and total labor becomes 5786.37. A 25% ground requirement uses 801.56 ground hours and 5486.15 total hours. These are separate assumption tests, not measured detection rates. Thus zero baseline ground screening depends on checklist equivalence and ample resources; operators and responders remain necessary.

Protocol	Rate (%)	Total labor (h)	Ground (h)	Flight (h)
Equivalent checklist	57.26	5286.11	0.00	144.34
Drone efficiency 75%	57.26	5786.37	1731.55	104.27
Drone efficiency 50%	57.26	6011.98	2964.54	33.35
Ground minimum 10%	57.26	5366.13	320.63	129.91
Ground minimum 25%	57.26	5486.15	801.56	108.26

Protocol changes in Table 11 restore ground screening while the ample-budget completion rate remains geographically limited.

Table 12 changes only grid size while retaining the area-based task density.

Table 12: Spatial discretization sensitivity with task density, resource budgets and response rules held fixed. The 5, 10 and 15 kilometer grids produce 1554, 599 and 392 connected inspection units, respectively. Their completion rates are 56.39%, 57.26% and 59.99%, and minimum personnel work is 5279.44, 5286.11 and 5340.08 hours. Total prescribed checks remain proportional to park area, so grid refinement does not create extra aggregate demand. Representative locations and response qualification change, particularly near the two-hour boundary. The finer result is 0.87 percentage points below the baseline and the coarser result is 2.73 above it. These differences measure approximation sensitivity rather than sampling uncertainty or a statistical confidence interval.

Grid side (km)	Units	Rate (%)	Total labor (h)
5	1554	56.39	5279.44
10	599	57.26	5286.11
15	392	59.99	5340.08

Representative locations and response eligibility cause discretization sensitivity; this is not a statistical confidence interval.

7.2 Weight sensitivity

Reoptimization keeps other conditions fixed. Entropy measures spatial differences; CRITIC uses variation and indicator overlap. Table 13 distinguishes completion rates under each weight choice from evaluation using the original weights.

The separate scarcity trial allows 3600 personnel-hours, including 2880 readiness hours, and 1200 flight-hours. Up to 9 units change completion relative to its entropy solution, while rates evaluated with the original weights span only 29.86–29.88%. Deployment responds to priorities, but the completion-rate effect is small in these tests.

Perturbing specially protected category preferences by $\pm 20\%$, with reciprocal comparisons and recalculated weights, preserves matrix consistency and base deployment. Tight-budget completion-rate losses stay below 0.002 percentage points. All cases pass independent capacity, eligibility, regional-floor and budget checks; robustness applies to these tested choices.

Table 13: Responsibility-weight sensitivity and evaluation using a common reference rule. Entropy, equal components, animal emphasis and CRITIC assign different animal--habitat shares while preserving routes, budgets and checklist requirements. The own-weight column scores each optimized plan under its chosen rule; original-weight evaluation applies the baseline entropy rule to all plans. At the ample baseline, deployment is identical and original-weight completion remains 57.26%, despite changes in the scoring scale. The tight-budget column uses original weights and the separate 3600-hour labor constraint; results span 29.86% to 29.88%. This paired evaluation prevents a changed definition of importance from masquerading as improved service. Weight choices remain management scenarios rather than measured ecological loss shares.

Weight choice	Animal (%)	Own weights	Original weights	Tight budget
Entropy baseline	64.56	57.26	57.26	29.88
Equal components	50.00	53.18	57.26	29.87
Animal emphasis	90.00	64.40	57.26	29.88
CRITIC alternative	44.80	51.72	57.26	29.86

8. Strengths, Limitations and Practical Use

We assess computational strengths and the evidence needed for operational use.

8.1 Strengths

The task-based framework separates budget scarcity from response access. Shared personnel and distinct flight constraints make resource use explicit; inverse workloads explain seasonal demand, with independent checks enabling reproducibility.

8.2 Weaknesses and Limitations

Table 14 links model limitations to their consequences and the field evidence needed for operational use.

Table 14: Limits that affect operational interpretation of the protection-planning model. Historical and incomplete object records restrict whose responsibility can be quantified; checklist and cost assumptions restrict the meaning of service completion. Mapped roads do not establish legal or seasonal permissions, and pooled monthly hours do not ensure qualified crews at every time. The 17-point maintenance sample cannot identify all current facilities or repair tasks. These limits explain why the computed 57.26% optimum and 57 or 62 personnel equivalents cannot certify sufficient ecological protection or actual park staffing. The final column identifies corresponding evidence needs. Independent mathematical checks address feasibility of the specified model, while incident, survey, habitat and field-work records are needed to validate real outcomes.

Limit	Consequence and data needed
Historical / missing objects	Current priority-species locations and surveys are needed; six measured groups do not certify all wildlife.
Service proxies and assumptions	Calibrate check frequency, category priorities, fire exposure and ground/drone checklist equivalence.
Maps and permissions	Verify roads, barriers, seasons and emergency access; a mapped route is not confirmed permission.
Monthly pooled work	Check dates, qualifications, integer visits and concurrent incidents before calling a roster feasible.
Incomplete maintenance inventory	Update borehole status, tasks and repair duration; the 17-point sample cannot define all current work.
Service versus ecological outcomes	Validate loss, habitat and population outcomes independently; optimal service does not prove adequate protection.

Before deployment, verify access and task effectiveness, update records, retain safeguards for missing objects, and validate ecological outcomes independently.

9. Adaptation to Parks on Other Continents

We examine Chitwan National Park in Asia and Yellowstone National Park in North America, rebuilding local geography and testing how priorities affect deployment.

9.1 Shared structure and local data

Priority setting, task capacity, response gates, readiness budgets and workforce conversion can remain. Species, habitats, seasonal routes, task protocols, permissions and operating costs must be rebuilt locally. Etosha's animal weights and thirty-drone assumption are not transferred.

Figure 7 locates the spatial inputs; Table 15 summarizes their published boundaries, roads and World-Cover 2021 classification. It assigns land-area weights $w_j = A_j^{\text{land}} / \sum_k A_k^{\text{land}}$ and $C_j = 8A_j^{\text{land}}/100$. Drones

are set to zero because authorization and task equivalence are not calibrated. These choices test spatial service, not a complete evaluation of species protection. The park trials impose the 15% floor in each reachable cell because reporting regions are unavailable; Etosha retains regional floors. The safeguard scale is locally modified.

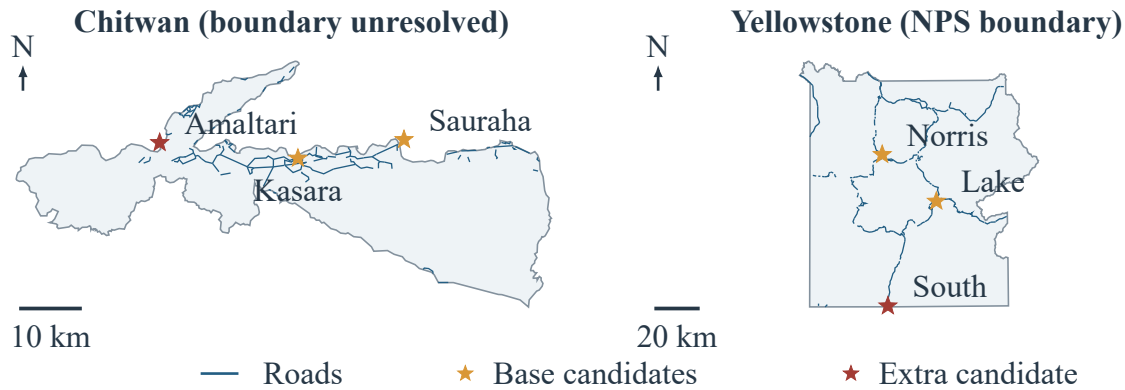


Figure 7: Geographic inputs and response candidates for the two cross-continent trials. Chitwan uses a public protected-area polygon, roads and geographic anchors; Yellowstone uses the NPS boundary, public roads and station locations. Stars identify baseline and additional candidates, rather than staffing assignments. Metric projections support travel calculations, while local speeds, task times and permissions remain assumptions. Chitwan's projected polygon covers 1214.06 square kilometers, exceeding the official 952.63 by 27.44%, so its trial remains uncalibrated. Yellowstone's mapped inputs support local terrestrial-service calculations. Neither panel transfers Etosha species weights, assumes its thirty-drone fleet, or establishes continuous monitoring, animal distributions or staffing requirements.

Table 15: Spatial inputs for Chitwan and Yellowstone trials. Land service areas exclude sampled permanent water, and forest shares come from WorldCover 2021 classification sampled at 100 meters. Road lengths and planning cells support route and response calculations. The Chitwan polygon supplies 1175.65 square kilometers of land service area and 1376 one-kilometer cells; the Yellowstone NPS boundary supplies 8470.07 and 1520 cells of 2.5 kilometers. These data rebuild local geography without calibrating animal protection. Chitwan's boundary-area inconsistency remains explicit. Areas and grids make scores local task measures, so these scores cannot rank ecological condition or transfer staffing between parks.

Input	Chitwan polygon	Yellowstone NPS
Land service area (km ²)	1175.65	8470.07
Forest share (%)	91.11	62.93
Mapped park roads (km)	248.29	619.60
Units / grid side	1376 / 1 km	1520 / 2.5 km

Boundary restriction. Chitwan's 1214.06 km² polygon exceeds the official 952.63 by 27.44%. Its 1213.00 km² geodesic area matches source GIS; projection accounts for only 0.087%. The official 2020 GIS report also gives 952.63 [16]. The boundary/reporting convention remains unresolved. Yellowstone uses NPS geography [13].

9.2 Conditional trials and expected deployment changes

For each park, fix its target at 90% of its own two-post, 30 km/h response-limited ceiling:

$$P_{\text{geo}}^{(p)} = 100 \sum_j w_j^{(p)} r_j^{(p)}, \quad (9.1)$$

$$\eta^{(p)} = 0.90 P_{\text{geo,base}}^{(p)}.$$

Table 16 retains fixed targets. Two posts reserve 960 personnel-hours; a third adds 480. Local specialist tasks remain unmeasured.

Table 16: Conditional land-inspection trials under local operating rules. Each park's target is fixed at 90% of its two-post, 30 kilometer-per-hour response ceiling, rather than whole-park 90%. Base-line Chitwan and Yellowstone require 1085.15 and 1541.63 personnel-hours, or 10 and 13 task equivalents. At 20 kilometers per hour, both fixed targets exceed the geographic ceilings and are infeasible. Adding a third post expands response eligibility but adds 480 readiness hours, increasing work at the original target. Higher speeds are sensitivity assumptions rather than driving recommendations. Chitwan remains a public-polygon trial, and neither park's result includes unmeasured community, visitor or concurrent tasks.

Case	Setting	Ceiling	Target	Labor (h)	People
C0	Two posts, 30 km/h	37.57	33.81	1085.15	10
C1	20 km/h	30.66	33.81	Infeasible	–
C2	Third post added	45.58	33.81	1550.39	13
C3	40 km/h	42.12	33.81	1069.56	9
Y0	Two posts, 30 km/h	18.72	16.85	1541.63	13
Y1	20 km/h	12.02	16.85	Infeasible	–
Y2	Third post added	21.60	16.85	1973.23	17
Y3	40 km/h	24.48	16.85	1467.97	13

Higher speed is a sensitivity assumption, not a driving recommendation.

Calibrate Chitwan's forest, river and community tasks, and Yellowstone's visitor and winter tasks; closures do not establish emergency permissions [14].

Half-grid tests change caps by -0.13 and -0.20 points; finer land-cover sampling changes forest shares by -0.002 and -0.056 points. Ten numerical cases pass; Chitwan's boundary check fails. Staffing equivalents remain conditional.

9.3 A local historical-priority trial

Table 17 uses $w_j^{(\lambda)} = (1 - \lambda)w_j^{\text{land}} + \lambda w_j^{\text{hist}}$, preserving Y0's new-weight score. At a 50% historical share, 61 units change, labor is 1474.68 hours, and land-weight service falls from 16.85% to 15.12%. These are historical priority trade-offs, not ecological gains [17]. The adjacent appendix gives the full comparison.

Appendix: supplementary results

Table 17 supplies the complete historical-priority comparison discussed in Section 9.3. All road, task and response inputs are held fixed; the priorities and retained targets are declared scenario choices.

Table 17: Yellowstone historical-priority scenarios with fixed roads and tasks. Ten published 2022 wolf-pack activity files contain nine distinct geometries; their clipped union covers 4093.87 square kilometers. Normalized historical overlap is adjusted by existing cell land fraction and mixed with land-area responsibility using the wolf share. Each scenario retains Y0's score under new weights, then minimizes personnel work. At a 50% historical share, 61 units change service and work falls to 1474.68 hours, while original land-weight completion falls from 16.85% to 15.12%. This is a priority trade-off, not improved ecological protection. Historical activity extents are not current density, measured threat or official management weights.

Wolf share (%)	New-rule rate	Land-rule rate	Labor (h)	Changed units
0	16.85	16.85	1541.63	0
50	20.04	15.12	1474.68	61
75	21.64	13.04	1413.44	102

The following reproducibility entry calls the same allocation solver using saved real inputs. It reports the rate, minimum personnel-hours and flight-hours, without supplying synthetic observations. Its exact source is also included in the support archive.

Letter to IMMC

To the International Mission for Monitoring Conservation

Subject: A practical protection plan for Etosha and other large reserves

Dear IMMC Decision-Makers,

We recommend connecting monitoring to timely ground response. Our Etosha study shows that staff and drones can remain unused while important locations are too distant for effective follow-up.

Our approach divides the park into planning areas, uses historical animal surveys and burnable habitat to guide priorities, and calculates task and travel time. Routes and response posts determine which checks receive support. Drones can save walking time in suitable open locations, but people must still transport and operate them and follow up reported problems.

The planning baseline uses 60% of 295 reference employees, or 177 available people at 120 effective hours each, thirty drones and two-hour ground arrival. These assumptions do not describe verified current staffing or equipment.

The highest effective inspection completion rate under these conditions is 57.26%. The minimum-work month uses about 5286 personnel-hours, including readiness, and 144 flight-hours. Both are well below the resource limits. This rate combines task completion and ground-response eligibility using conservation-priority weights. Wildlife losses, population health and habitat conditions need separate evaluation.

Among three additional-post candidates tested within the same budgets, ENP5 raises the completion rate to 64.27%. It is a direction for field assessment, not a confirmed global optimum. Check routes, permissions, site conditions and response needs before adoption.

Seasonal work also needs preparation. Additional fire checks and borehole maintenance raise the modeled normal peak requirement to 57 people. The specified drought-maintenance scenario raises it to 62. Both fit within the 177-person reference, but cover only the selected daytime tasks, response range and historical maintenance sample.

Begin with a limited field trial: record travel, inspection, drone use and follow-up time; compare checklist completion by ground and drone-assisted teams; update the maintenance inventory; and verify access. Use these records to revise routes, response locations and task assumptions.

Turn the monthly plan into a calendar. Stagger borehole visits to avoid long service gaps, check daily capacity and retain qualified response staff. Thirty drones do not imply thirty simultaneous flights; operator, readiness and maintenance work all need time.

Trials in Chitwan and Yellowstone show how the structure can be adapted using local spatial records and operating assumptions. Chitwan's boundary disagrees with the official area, so its trial cannot support a calibrated staffing recommendation. Yellowstone requires information on visitor interactions and winter access. Neither trial establishes actual park staffing.

At Etosha, records come from different years, and some priority species, including black rhinoceros, lack detailed distributions. The model assumes occasional alerts rather than several serious simultaneous incidents. Specialist and night work may add personnel needs; a high inspection completion rate cannot resolve these gaps.

We propose three connected actions: validate field assumptions; examine response locations that could close important access gaps; and revise monthly assignments using verified records. Retain basic service in less prominent areas and dedicated safeguards for priority species. Measure progress through completed checks, reliable follow-up and independently observed wildlife and habitat outcomes.

Personnel, equipment and access should be planned together. The framework provides a transparent starting point, with decisions revised as reliable evidence improves.

Yours faithfully,

The Modeling Team

Team 510105201001220151

References

- [1] IM²C. *Protecting Wildlife at Scale*, 2026 problem statement and submission rules supplied to the team.
- [2] *Etosha aerial wildlife survey report*, 2015. Regional abundance source. Online source.
- [3] Namibia. *Nature Conservation Ordinance 4 of 1975*, consolidated legal categories. Online source.
- [4] ESA. *WorldCover 2021*, v200. Land-cover data and classification. Online source.
- [5] OpenStreetMap contributors / Geofabrik. Namibia snapshot, 3 October 2026; Nepal source records documented in the project. Online source.
- [6] Namibia Ministry of Environment and Tourism. *Fire Management Strategy*, 2016, Etosha operating windows. Online source.
- [7] Riddell, E. S., Kilian, W., Versfeld, W., and Kosoana, M. Groundwater stable isotope profile of the Etosha National Park, Namibia. *Koedoe*, 58(1), 2016. Table 1 records the 2013 sample. Online source.
- [8] Namibia MEFT. *Annual Report 2021/2022*. Artificial-water-point equipment evidence. Online source.
- [9] Dryad. Etosha study dataset, 2024; study-area dry/wet-season classification. Online source.
- [10] UNESCO World Heritage Centre. *Chitwan National Park* and associated conservation records. Online source.
- [11] Nepal DNPWC. *Chitwan National Park Management Plan 2013–2017* and official UNESCO report; boundary-area reference 952.63 km². Online source.
- [12] UNEP-WCMC. *WDPCA / Protected Planet*, October 2026, site 805 published polygon. Its area mismatch is retained as a failed calibration check. Online source.
- [13] US National Park Service. Yellowstone boundary, public roads and location datasets; source queries and fingerprints retained. Online source.
- [14] US National Park Service. Yellowstone visitor use, current conditions, and laws and policies. Online source.
- [15] OpenAI. *Codex*, AI coding assistant used in the October 2026 work session; GPT-6 model family, exact deployed version unavailable. Uses and verification are described in the AI-use report. Online source.
- [16] Nepal DNPWC. *GIS Database of Protected Areas*, 2020. Chitwan district-area table, p. 17. Official report.
- [17] US National Park Service. *Yellowstone wolf territories, 1995–2022*; 2022 95% MCP files. Official GIS archive.

Report on Use of AI

Tool and identification. OpenAI Codex, based on the GPT-6 model family, was used in this work session [15]. The exact deployed model version was not exposed in the session metadata. Python/SciPy and XeLaTeX performed numerical optimization and document compilation; those outputs were checked separately from generated prose.

Uses. The assistant helped explain existing formulas, distinguish sourced records from planning assumptions, revise resource parameters specified by the team, synchronize seasonal calculations, shorten chapter drafts, translate the integrated manuscript into English, draft the nontechnical letter and produce numbered figures and tables. Parallel assistants worked on bounded chapter, data-synchronization and visualization tasks.

Representative instructions. The team requested: (1) explain each stage and formula in the existing Question 2 document; (2) keep other parameters unchanged, make 60% of personnel available and use thirty drones, then shorten Question 2 to eight pages with simple terminology; (3) retrieve remote updates, revise Question 3 to six pages, revise Question 6 similarly, and integrate Questions 1–6, the template sections and a letter into an English paper of at most 24 pages. The team supplied the control number used here and subsequently requested a multi-species scope, correction of the two-stage objective and feasibility-based manuscript revision.

Verification. Reported Question 2–4 allocations were recomputed or checked against saved inputs and explicit constraints. The cross-park chapter uses the retrieved real-space results and retains its failed Chitwan boundary check. Scripts verify units, capacities, budgets, prohibitions, floors, totals, references, figure/table labels and final page count. The present revision includes renewed numerical checks and PDF layout review; the corresponding audit records are retained with the source. A computational check is not a validation of real ecological outcomes or a detailed roster.

Limits and responsibility. Generated wording and model choices require team review. The manuscript identifies historical data, scenario assumptions and uncalibrated inputs; it makes no claim that AI supplied current animal distributions, verified staffing inventories, ecological-loss effects or authorized access permissions. Sources are cited separately from the team's calculations. The team remains responsible for accuracy, originality and compliance with the competition's disclosure rules.

The uploaded template's obsolete year, example notation, sample identifiers and unfinished claims were replaced with the current problem, confirmed team number and verified results. The project retains source data, model inputs, allocation arrays, code, provenance and numerical/layout audit records for review.